

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Factoring Trinomials ($x^2 + bx + c$)

Example (Factor the trinomial):

$x^2 + 5x + 6$ — find two numbers that multiply to 6 and add to 5: that is 2 and 3. So
 $x^2 + 5x + 6 = (x + 2)(x + 3)$.

Directions: Factor by finding two numbers that multiply to c and add to b .

1. $x^2 + 7x + 10$

4. $x^2 + 2x - 8$

2. $x^2 + 8x + 15$

5. $x^2 - x - 12$

3. $x^2 - 5x + 6$

Challenge Problems

6. $x^2 - 2x - 15$ — factor, then name the values that make it 0.

Answer Key and Procedure

Procedure

1. List factor pairs of c .
2. Choose the pair that adds to b .
3. Write $(x + \text{first})(x + \text{second})$ and FOIL to check.

Answers

1. $(x + 2)(x + 5)$
2. $(x + 3)(x + 5)$
3. $(x - 2)(x - 3)$
4. $(x + 4)(x - 2)$
5. $(x - 4)(x + 3)$
6. $(x - 5)(x + 3)$; zeros $x = 5, -3$